

E. Jeffrey Samuels (192524412)

SET 13 – GEOGRAPHIC DATA ANALYSIS

Aim

To analyze geographic data using a map-style chart, scatter plot, and table in R.

Algorithm

1. Create the geographic dataset containing city, population, temperature, and elevation.
2. Assign simple latitude and longitude values for visualization.
3. Create a map-style plot showing city locations.
4. Create a scatter plot between population and average temperature.
5. Display the geographic data in a table.

R Program

```
library(ggplot2)
```

```
geo <- data.frame(
```

```
  City=c("City A","City B","City C"),
```

```
  Population=c(500000,700000,600000),
```

```
  Temperature=c(75,68,80),
```

```
  Elevation=c(1000,800,1200),
```

```
  Longitude=c(78,79,80),
```

```
  Latitude=c(13,14,12)
```

```
)
```

```
ggplot(geo, aes(Longitude, Latitude, size=Population, label=City)) +
```

```
  geom_point() +
```

```
  geom_text(vjust=-1) +
```

```
  labs(title="Geographic Distribution of Cities",
```

```
        x="Longitude", y="Latitude")
```

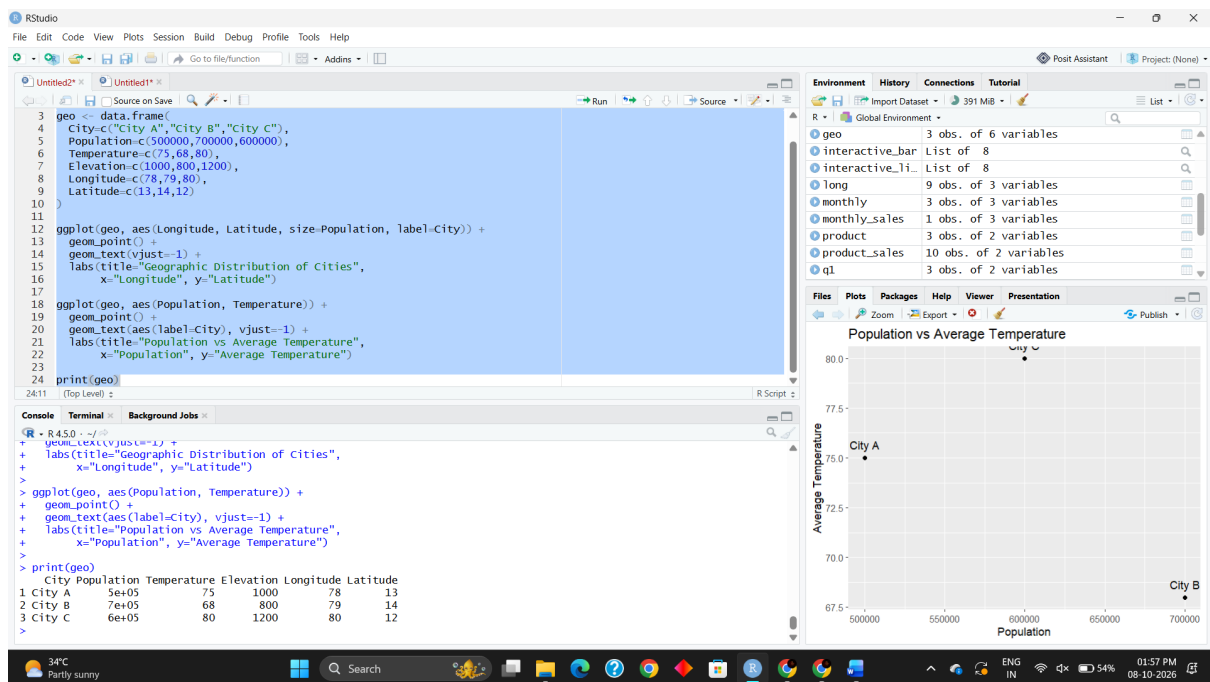
```
ggplot(geo, aes(Population, Temperature)) +
```

```
  geom_point() +
```

```
geom_text(aes(label=City), vjust=-1) +
labs(title="Population vs Average Temperature",
      x="Population", y="Average Temperature")
```

```
print(geo)
```

OUTPUT:



Result

The map-style plot successfully represents the geographic distribution of the cities. The scatter plot shows the relationship between population and average temperature, and the table displays the geographic information.